Views - Assignment 11

1) Create a view that shows all of the customers who have the highest ratings.

CREATE VIEW v_cust_max_rating AS

SELECT *

FROM customers

WHERE rating = (SELECT MAX(rating)

FROM customers);

SELECT cname

FROM v_cust_max_rating;

```
[d5_96525_Deven@(none)]>use sales;
Database changed
[d5_96525_Deven@sales]>CREATE VIEW v_cust_max_rating AS
    -> SELECT *
    -> FROM customers
    -> WHERE rating = (SELECT MAX(rating)
    ->          FROM customers);
Query OK, 0 rows affected (1.280 sec)

[d5_96525_Deven@sales]>SELECT cname
    -> FROM v_cust_max_rating;
+----------+
| cname    |
+----------+
| Grass    |
| Cisneros |
+----------+
2 rows in set (0.377 sec)

[d5_96525_Deven@sales]>
```

2) Create a view that shows the number of salespeople in each city.

CREATE VIEW v_count_salespeps AS
SELECT *

FROM salespeople;

SELECT city,COUNT(*)

FROM v_count_salespeps

GROUP BY city;

```
[d5_96525_Deven@sales]>SELECT city,COUNT(*)
    -> FROM v_count_salespeps
    -> GROUP BY city;
+-----------+----------+
| city      | COUNT(*) |
+-----------+----------+
| London    |        2 |
| San Jose  |        1 |
| Barcelona |        1 |
| New York  |        1 |
+-----------+----------+
4 rows in set (0.007 sec)
```

3) Create a view that shows the average and total orders for each salesperson after his or her name. Assume all names are unique.

CREATE VIEW v_sales_stats AS

SELECT s.sname, AVG(o.amt), COUNT(o.amt)

FROM salespeople s

JOIN orders o ON s.snum = o.snum

GROUP BY s.snum, s.sname;

SELECT * FROM v_sales_stats;

```
[d5_96525_Deven@sales]>SELECT * FROM v_sales_stats;
+---------+-------------+--------------+
| sname   | AVG(o.amt)  | COUNT(o.amt) |
+---------+-------------+--------------+
| Rifkin  |  558.425000 |            2 |
| Peel    | 5127.356667 |            3 |
| Motika  | 1900.100000 |            1 |
| Serres  | 1848.716667 |            3 |
| Axelrod | 1713.230000 |            1 |
+---------+-------------+--------------+
5 rows in set (0.011 sec)
```

4) Create a view that shows each salesperson with multiple customers. Changing Values through Views.

CREATE VIEW v_multicust_salespep AS

SELECT sname

FROM salespeople

WHERE snum IN (SELECT snum

FROM customers

GROUP BY snum

HAVING count(snum)>1);

SELECT *

FROM v_multicust_salespep;

```
[d5_96525_Deven@sales]>CREATE VIEW v_multicust_salespep AS
    -> SELECT sname
    -> FROM salespeople
    -> WHERE snum IN (  SELECT snum
    ->          FROM customers
    ->          GROUP BY snum
    ->          HAVING count(snum)>1);
Query OK, 0 rows affected (0.068 sec)

[d5_96525_Deven@sales]>SELECT *
    -> FROM v_multicust_salespep;
+--------+
| sname  |
+--------+
| Peel   |
| Serres |
+--------+
2 rows in set (0.123 sec)
```

5) Which of these views are updateable (will allow DML operations)?

#1 Create View Dailyorders as Select Distinct cnum, snum, onum, odate from Orders;

Create View Dailyorders as

Select Distinct cnum, snum, onum, odate

from Orders;

INSERT INTO Dailyorders VALUES (1,2,3,2003-02-06);

```
[d5_96525_Deven@sales]> INSERT INTO Dailyorders VALUES (1,2,3,2003-02-06);
ERROR 1471 (HY000): The target table Dailyorders of the INSERT is not insertable-into
[d5_96525_Deven@sales]>
```

#2 View Custotals as Select cname, Sum (amt) Sum_Amt from Orders, Customers where Orders.cnum=Customers.cnum Group by cname;

Create View Custotals as

Select cname, SUM(amt) AS Sum_Amt

from Orders, Customers

where Orders.cnum=Customers.cnum

Group by cname;

INSERT INTO Custotals VALUES ('DEVEN',45876);

```
[d5_96525_Deven@sales]>INSERT INTO Custotals VALUES ('DEVEN',45876);
ERROR 1471 (HY000): The target table Custotals of the INSERT is not insertable-into
[d5_96525_Deven@sales]>
```

#3 Create view Thirdorders as Select * from Dailyorders where odate=’1990-10-03’;

Create view Thirdorders as

Select *

from Dailyorders

where odate='1990-10-03';

INSERT INTO Thirdorders VALUES (1,2,3,1990-10-03);

```
[d5_96525_Deven@sales]>Create view Thirdorders as
    -> Select *
    -> from Dailyorders
    -> where odate='1990-10-03';
Query OK, 0 rows affected (0.161 sec)

[d5_96525_Deven@sales]>INSERT INTO Thirdorders VALUES (1,2,3,1990-10-03);
ERROR 1471 (HY000): The target table Thirdorders of the INSERT is not insertable-into
[d5_96525_Deven@sales]>
```

#4 Create view Nullcities as Select snum, sname, city from Salespeople where city is NULL OR sname BETWEEN ‘A’ and ‘MZ’;

Create view Nullcities as

Select snum, sname, city

from Salespeople

where city is NULL

OR sname BETWEEN 'A' and 'MZ';

INSERT INTO Nullcities VALUES (43,'DEVEN','London');

```
[d5_96525_Deven@sales]>Create view Nullcities as
    -> Select snum, sname, city
    -> from Salespeople
    -> where city is NULL
    -> OR sname BETWEEN 'A' and 'MZ';
Query OK, 0 rows affected (0.176 sec)

[d5_96525_Deven@sales]>INSERT INTO Nullcities VALUES (43,'DEVEN','London');
Query OK, 1 row affected (0.186 sec)

[d5_96525_Deven@sales]>
```

6) Create a view of the Salespeople table called Commissions. This view will include only the snum and comm fields. Through this view, someone could enter or change commissions, but only to values between .10 and .20.

CREATE view Commissions AS

SELECT snum,comm

FROM salespeople

WHERE comm BETWEEN 0.10 AND 0.20;

INSERT INTO Commissions VALUES (1,0.15);

```
[d5_96525_Deven@sales]>CREATE view Commissions AS
    -> SELECT snum,comm
    -> FROM salespeople
    -> WHERE comm BETWEEN 0.10 AND 0.20;
Query OK, 0 rows affected (0.160 sec)

[d5_96525_Deven@sales]>INSERT INTO Commissions VALUES (1,0.15);
Query OK, 1 row affected (0.177 sec)

[d5_96525_Deven@sales]>table salespeople;
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1001 | Peel    | London    | 0.12 |
| 1002 | Serres  | San Jose  | 0.13 |
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
|   43 | DEVEN   | London    | NULL |
|    1 | NULL    | NULL      | 0.15 |
+------+---------+-----------+------+
7 rows in set (0.006 sec)

[d5_96525_Deven@sales]>
```